

Lecture 12 Image Reconstruction

(Supervised Deep Learning Methods)

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SIST Building-3 420

Outline

- Image Reconstruction Formulation
- Typical Deep Learning Strategies
 - Learning Prior in Image Domain
 - Learning Prior in Sensor Domain
 - Learning Prior in-between Domain
 - Learning Transform in-between Domain

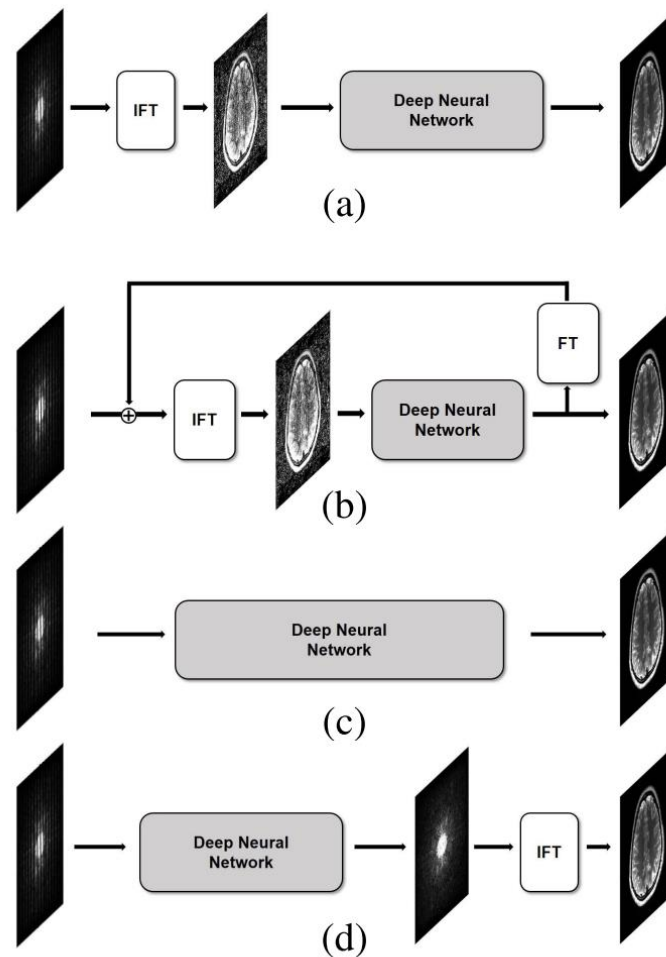


Fig. 1. Deep learning frameworks for accelerated MRI: (a) image domain learning [6], [7], (b) cascaded network [4], [5], [9], (c) AUTOMAP [10], and (d) the proposed k -space learning. IFT: Inverse Fourier transform.

Image Reconstruction Formulation

- Medical image reconstruction can often be formulated as the following mathematical problem:

$$y = \Phi x + \eta$$

- where Φ is a physical system modeling the image acquisition process. Φ can be a linear operator or nonlinear operator that depends on the specific imaging modality.
- x is the unknown image to be reconstructed.
- y is the measured data that might be corrupted by noise η with known or partially known noise statistics, e.g., Gaussian, Laplacian, Poisson, Rayleigh, etc.



Image Reconstruction Formulation

- The inverse problem is in general challenging to solve due to the large-scale and ill-posed nature of the problem in practice.
- To solve the inverse problem, it is common practice to consider the following optimization problem:

$$\min_{\mathbf{x}} \mathcal{L}(\mathbf{x}) = F(\Phi \mathbf{x}, \mathbf{y}) + \lambda \Psi(\mathbf{W}, \mathbf{u})$$

- The solution $\hat{\mathbf{x}} = \mathit{argmin} \mathcal{L}(\mathbf{x})$ is an approximate solution to the inverse problem.
- The term $F(\Phi \mathbf{x}, \mathbf{y})$ is the data fidelity term that measures the consistency of the approximate solution. $\Psi(\mathbf{W}, \mathbf{u})$ is the regularization term encoding the prior knowledge on the image to be reconstructed.

Difference and Similarity of Image Reconstruction and other Image processing

$$\min_{\mathbf{x}} \mathcal{L}(\mathbf{x}) = F(\Phi \mathbf{x}, \mathbf{y}) + \lambda \Psi(\mathbf{W}, \mathbf{u})$$

- Major difference : due to Φ , $\mathbf{x}$ and $\mathbf{y}$ are in different domain, for other image processing problems, $\mathbf{x}$ and $\mathbf{y}$ are both in image domain.
- Similarity: in most image reconstruction case, the domain transfer forward model is known, so it is possible to formulate loss of the $\mathbf{x}$ and $\mathbf{y}$ in the same domain.



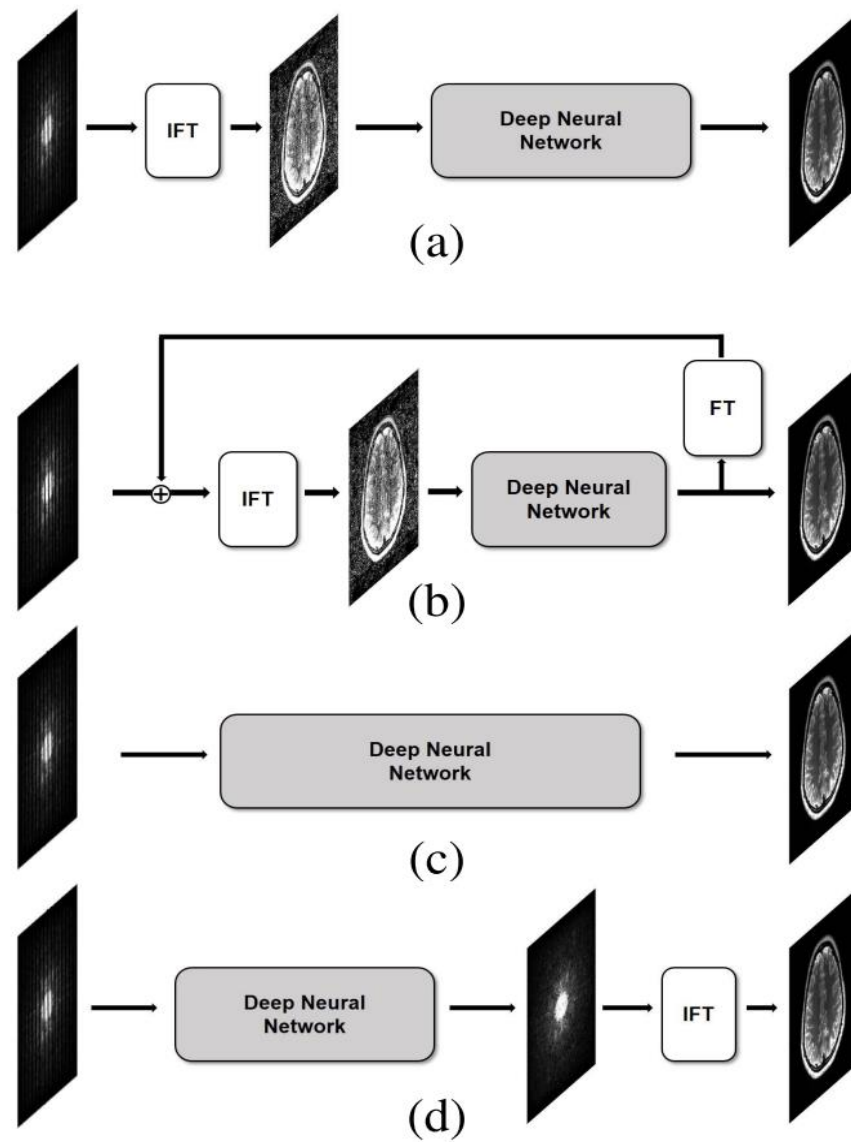


Fig. 1. Deep learning frameworks for accelerated MRI: **(a)** image domain learning [6], [7], **(b)** cascaded network [4], [5], [9], **(c)** AUTOMAP [10], and **(d)** the proposed k -space learning. IFT: Inverse Fourier transform.

Solve Pb in image domain

- [1] Zhicheng Zhang, Xiaokun Liang , Xu Dong, Yaoqin Xie, and Guohua Cao: A Sparse-View CT Reconstruction Method Based on Combination of DenseNet and Deconvolution, TMI 2018
- [2] Yoseb Han and Jong Chul Ye, Framing U-Net via Deep Convolutional Framelets: Application to Sparse-View CT, TMI 2018
- [3] Tiancheng Shen, Xia Li, Zhisheng Zhong, Jianlong Wu and Zhouchen Lin: Lin2R2-Net: Recurrent and Recursive Network for Sparse-View CT Artifacts Removal, MICCAI 2019

PB Formulation 1: Solve Pb in image domain

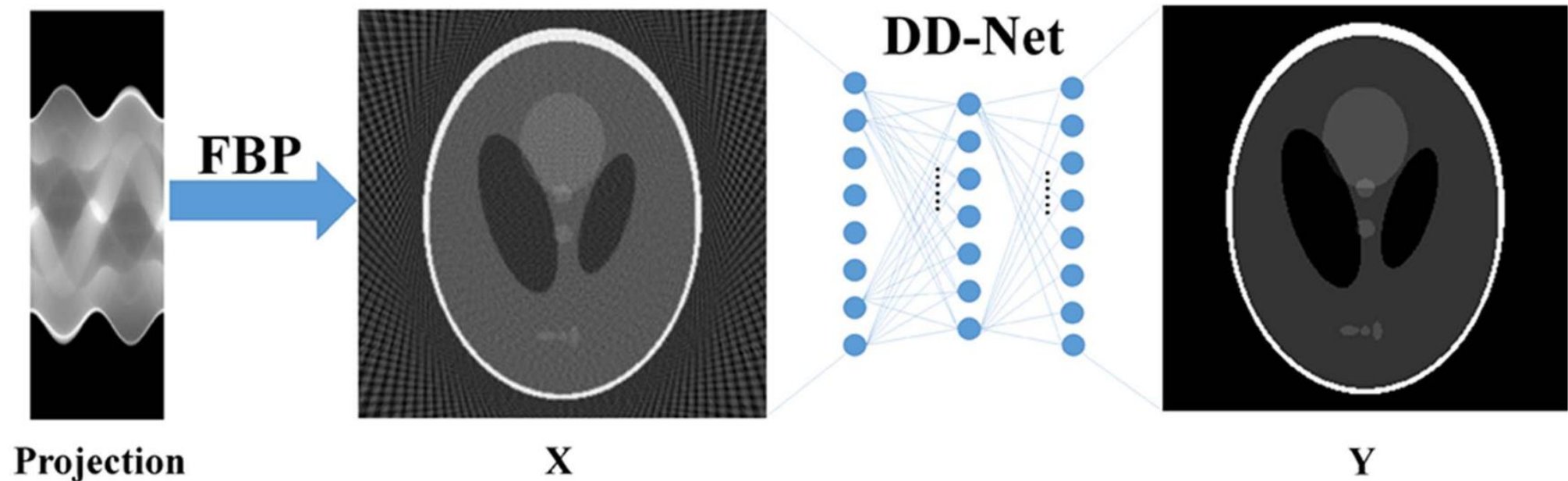
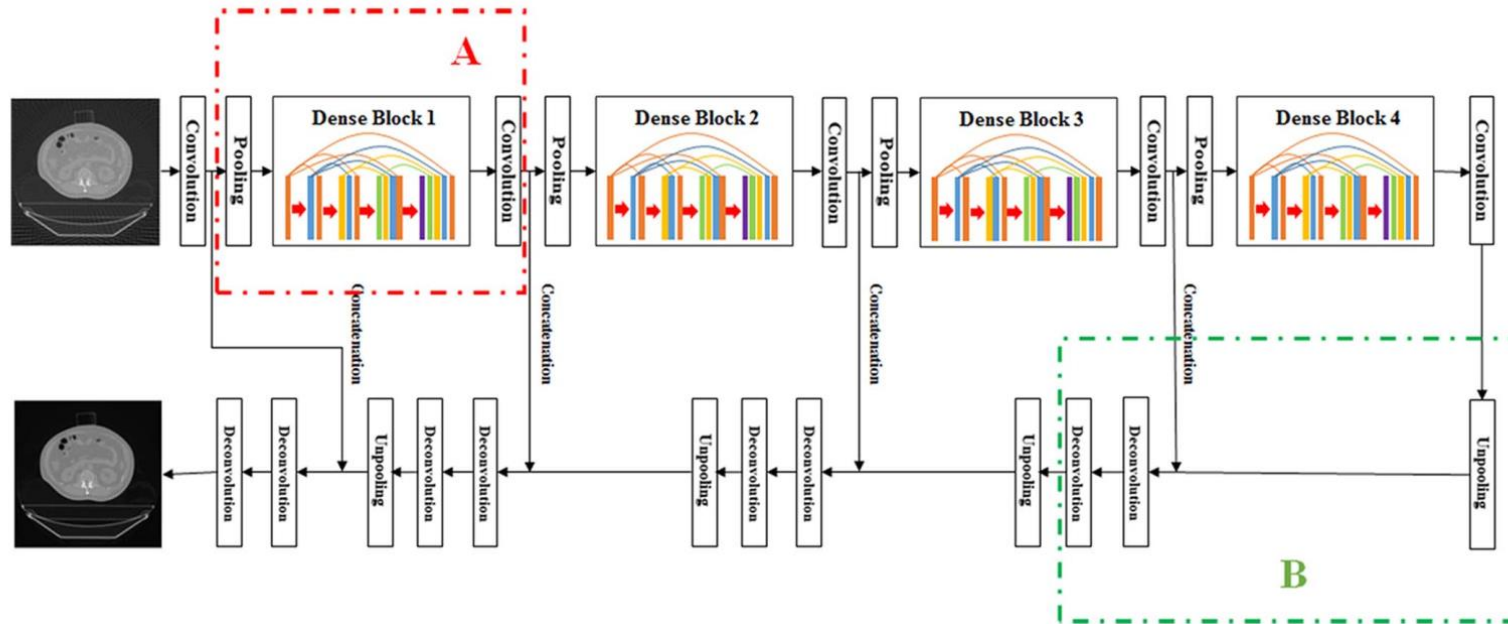


Fig. 1. Framework of the proposed sparse-view CT reconstruction method. Projection data from sparse-view CT are first reconstructed using FBP, and then the FBP reconstructed image is improved using a neural network (DD-Net).

PB Formulation 1: Solve Pb in image domain



[1] Zhicheng Zhang, Xiaokun Liang, Xu Dong, Yaoqin Xie, and Guohua Cao: A Sparse-View CT Reconstruction Method Based on Combination of DenseNet and Deconvolution, TMI 2018

Fig. 2. The corresponding network architecture of the DD-Net.

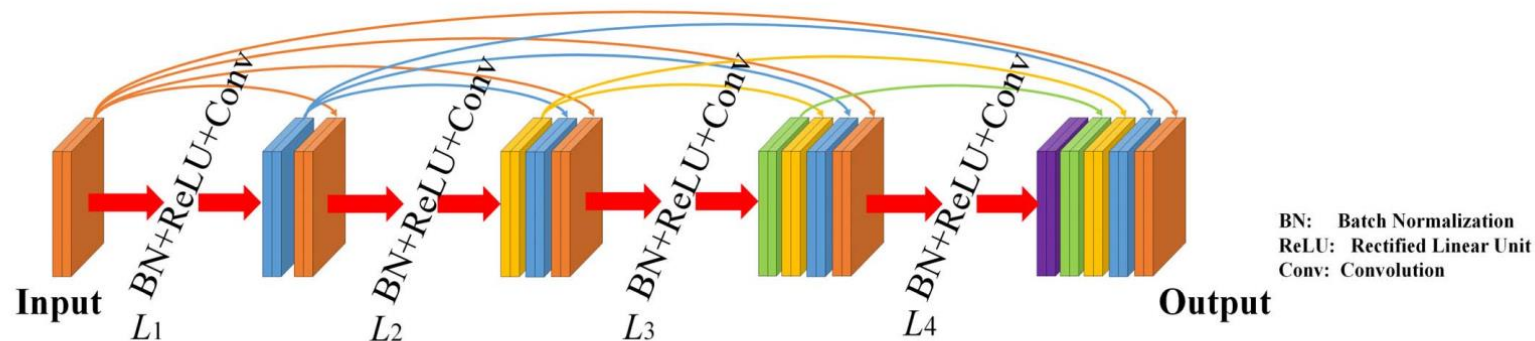


Fig. 3. A dense block with 4 L layers. Each L layer concatenates all preceding features together as its input.

PB Formulation 1: Solve Pb in image domain

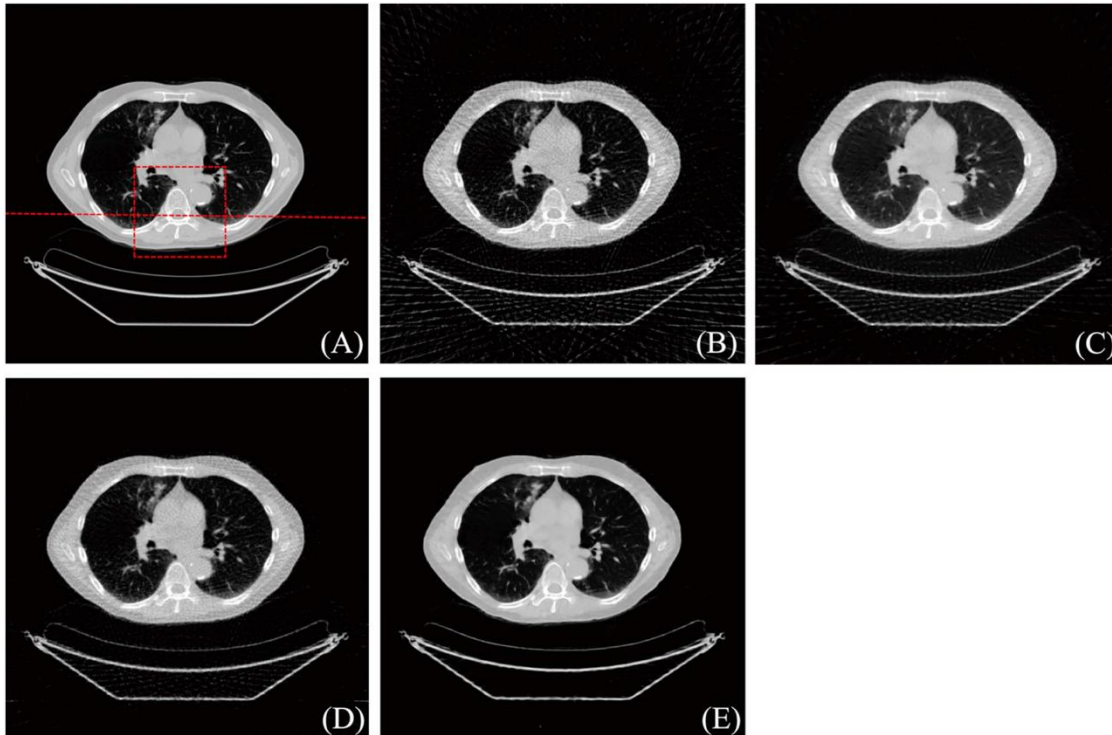


Fig. 5. Reconstruction results from the chest CT dataset. (A) Original image, (B) FBP, (C) PWLS-TGV, (D) R-NLTV, (E) DD-Net. Display window was set at $[0, 0.029] \text{ mm}^{-1}$.

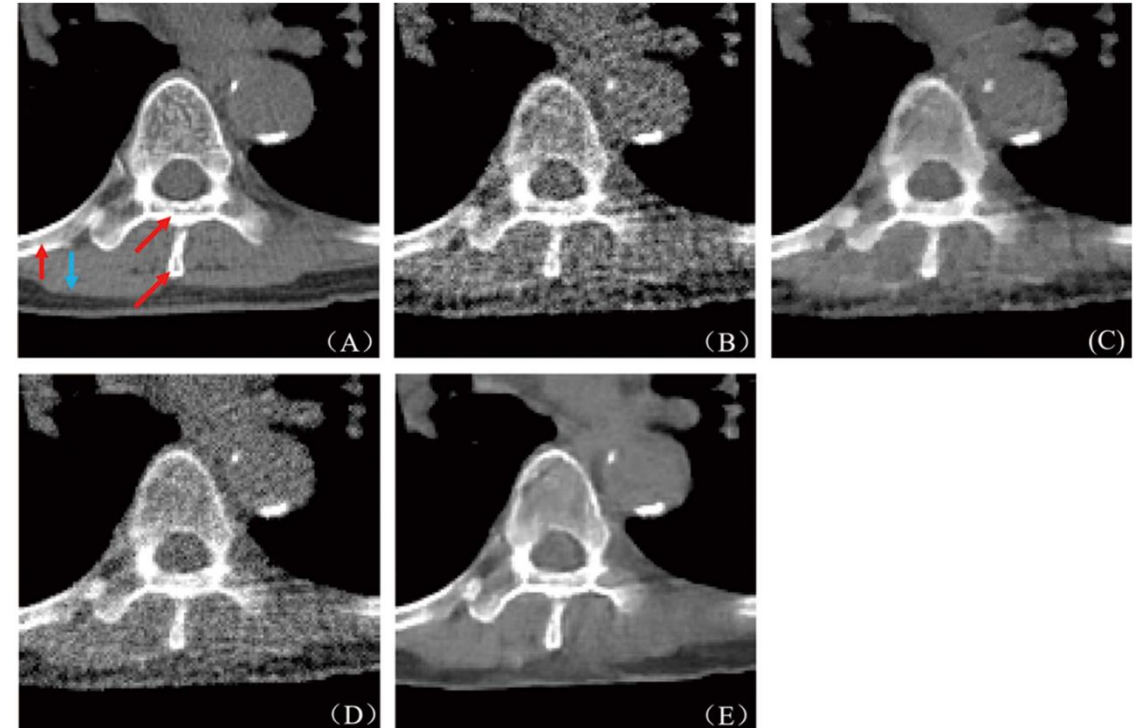


Fig. 6. The zoomed regions marked by the red box in Figure 5 (A). (A) Original image, (B) FBP, (C) PWLS-TGV, (D) R-NLTV, (E) DD-Net. The red arrows show the locations of the small features, and the blue arrow indicates the boundary between two soft tissues. Display window was set at $[0.00164, 0.0287] \text{ mm}^{-1}$.

PB Formulation 1: Solve Pb in image domain

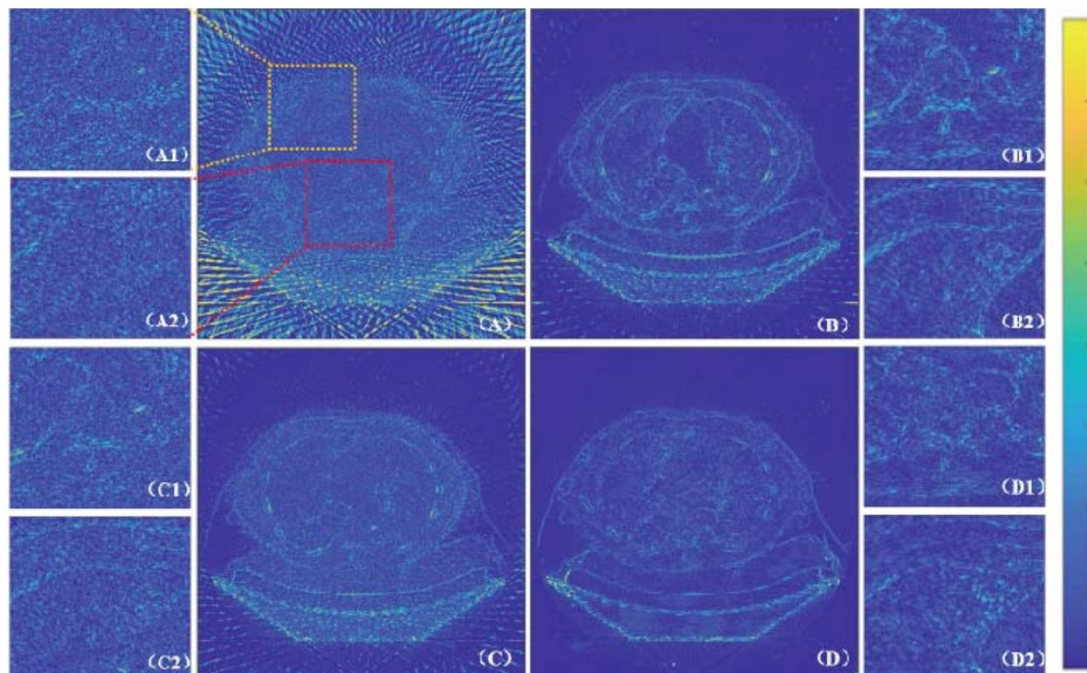


Fig. 8. The absolute difference images between the original CT image and the CT images reconstructed from the different algorithms: (A) FBP, (A1) enlarged area marked by orange box and (A2) enlarged area marked by red box; (B) PWLS-TGV, (B1) and (B2) are corresponding enlarged areas; (C) R-NLTV, (C1) and (C2) are corresponding enlarged areas; and (D) DD-Net, (D1) and (D2) are corresponding enlarged areas. Display window was set at $[0, 0.008] \text{ mm}^{-1}$. A larger view of the figure is available in the supplemental document.

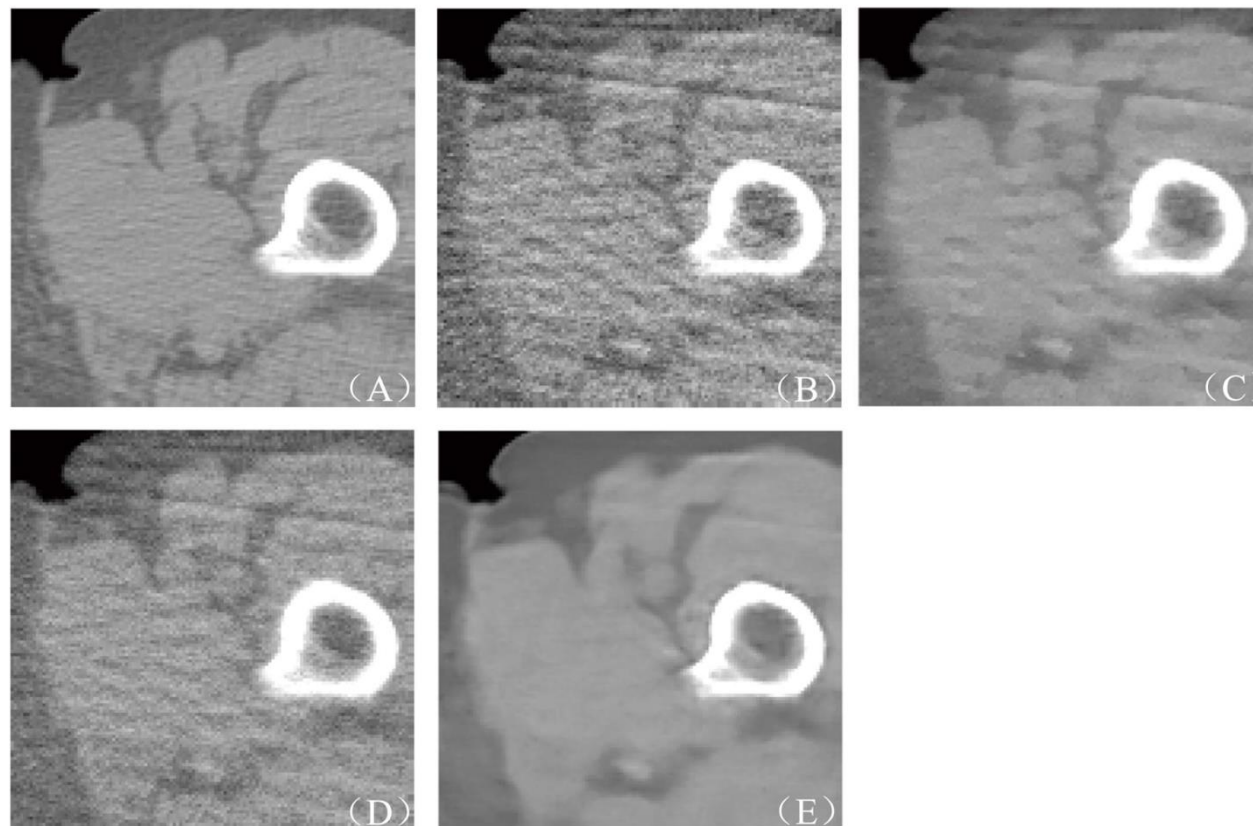


Fig. 10. The zoomed region marked by the blue box in Figure 9 (A). (A) Original image, (B) FBP, (C) PWLS-TGV, (D) R-NLTV, (E) DD-Net. Display window was set at $[0.012, 0.029] \text{ mm}^{-1}$.



PB Formulation 1: Solve Pb in image domain

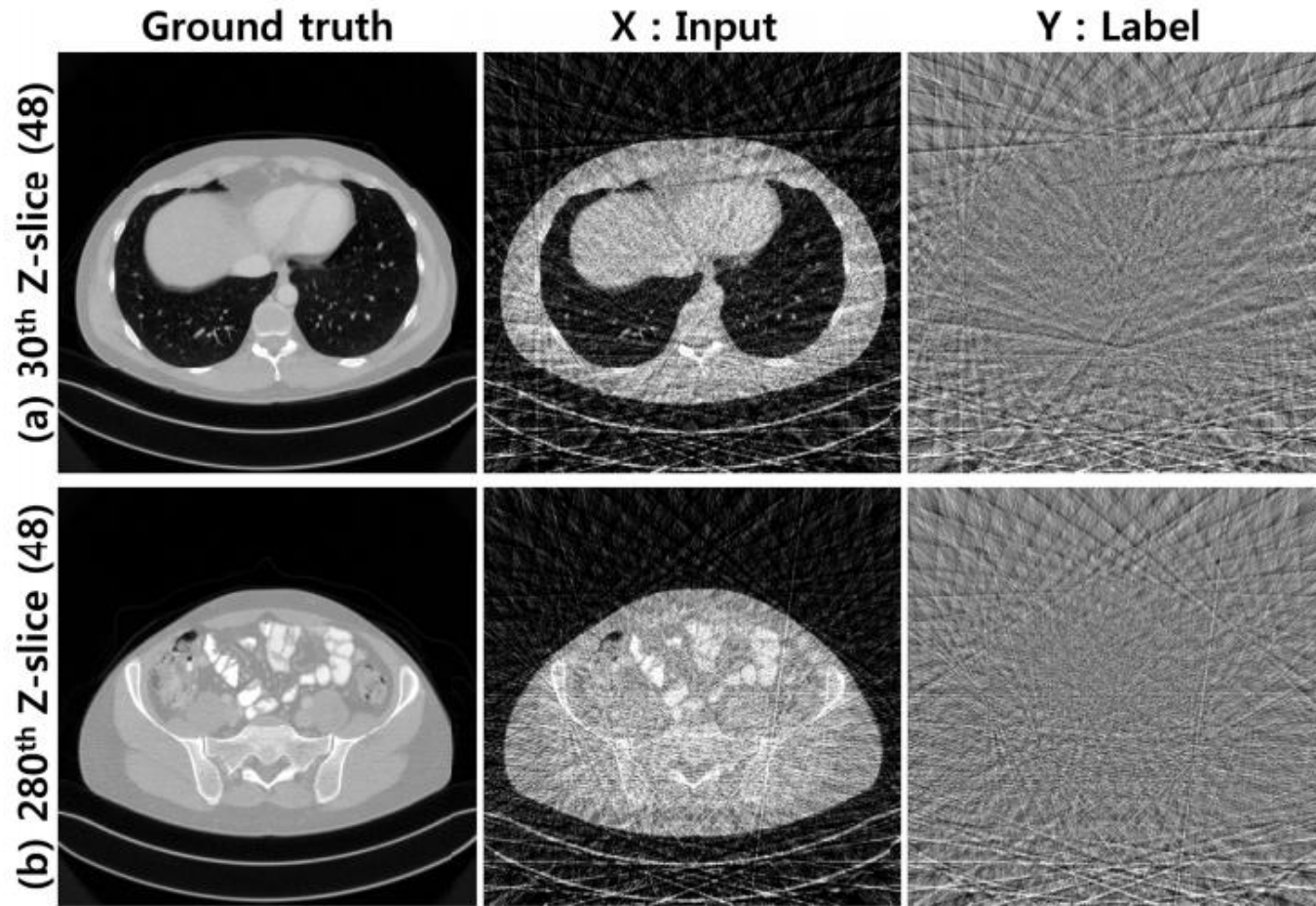
TABLE IV

QUANTITATIVE RESULTS FROM DIFFERENT RECONSTRUCTION
METHODS USING DIFFERENT NUMBERS OF PROJECTIONS

Method		60	72	90	120	180
RMSE (mm^{-1})	FBP	0.0032	0.0027	0.0023	0.0019	0.0015
	PWLS-TGV	0.0015	0.0013	0.0011	0.00092	0.00073
	R-NLTV	0.0017	0.0015	0.0013	0.0011	0.00086
	DD-Net	0.0016	0.0012	0.00089	0.00067	0.00057
SSIM	FBP	0.36	0.38	0.42	0.47	0.57
	PWLS-TGV	0.80	0.81	0.84	0.86	0.89
	R-NLTV	0.71	0.72	0.75	0.79	0.83
	DD-Net	0.80	0.86	0.90	0.93	0.94



PB Formulation 1: Solve Pb in image domain



[2] Yoseb Han and Jong Chul Ye, Framing U-Net via Deep Convolutional Framelets: Application to Sparse-View CT, TMI 2018

Fig. 1. CT streaking artifact patterns in the reconstruction images from 48 projection views.

PB Formulation 1: Solve Pb in image domain

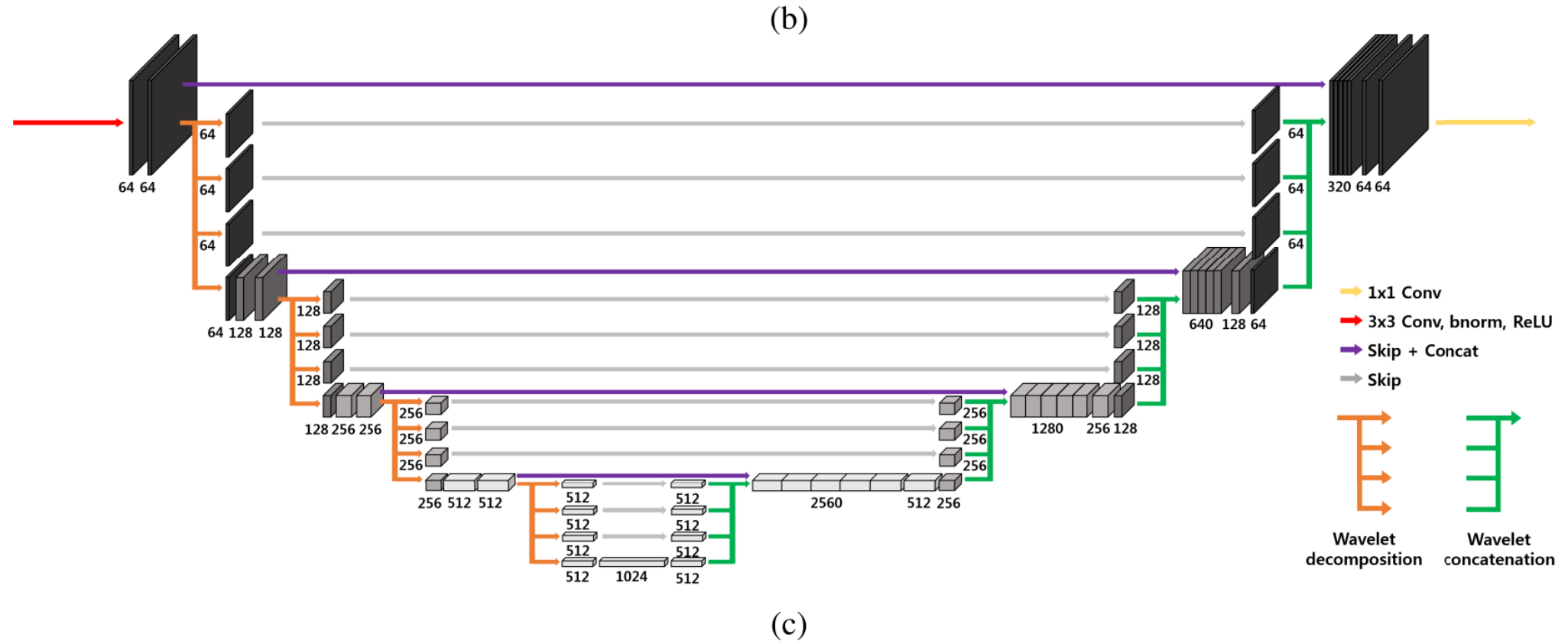


Fig. 4. 2-D U-Net architecture and its variants. (a) Standard U-Net, (b) dual frame U-Net, and (c) tight frame U-Net with concatenation.

[2] Yoseb Han and Jong Chul Ye, Framing U-Net via Deep Convolutional Framelets: Application to Sparse-View CT, TMI 2018

PB Formulation 1: Solve Pb in image domain

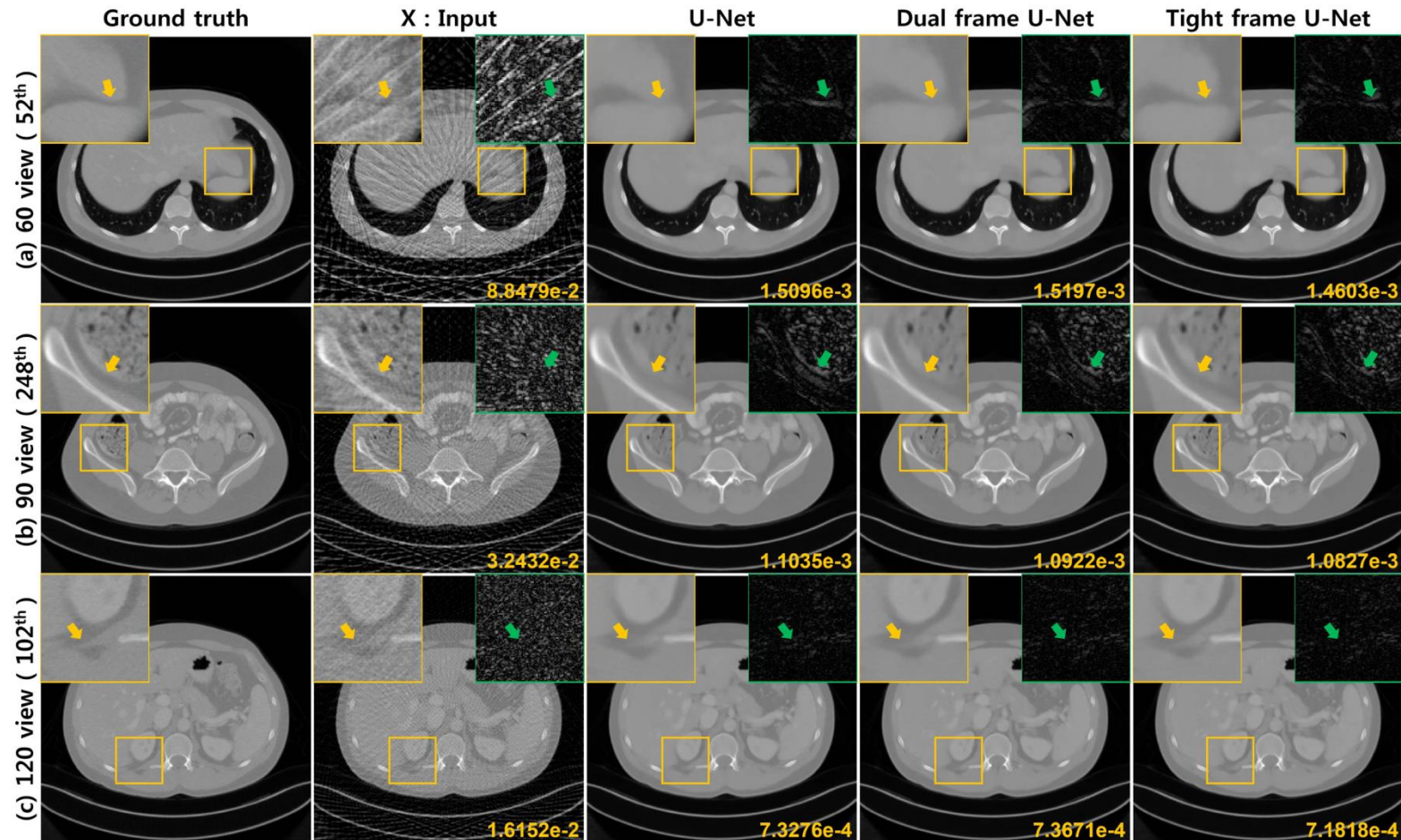


Fig. 5. Reconstruction results by original, dual frame and tight frame U-Nets at various sparse view reconstruction. Yellow and green boxes illustrate the enlarged view and the difference images, respectively. The number written to the images is the NMSE value.

[2] Yoseb Han and Jong Chul Ye, Framing U-Net via Deep Convolutional Framelets: Application to Sparse-View CT, TMI 2018

PB Formulation 1: Solve Pb in image domain

TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT METHODS

PSNR [dB] (whole image area)	60 views (x12)	90 views (x8)	120 views (x6)	180 views (x4)	240 views (x3)	360 views (x2)
FBP	22.2787	25.3070	27.4840	31.8291	35.0178	40.6892
U-Net	38.8122	40.4124	41.9699	43.0939	44.3413	45.2366
Dual frame U-Net	38.7871	40.4021	41.9397	43.0795	44.3211	45.2816
Tight frame U-Net	38.9218	40.5091	42.0457	43.1800	44.3952	45.2552
PSNR [dB] (within body)	60 views (x12)	90 views (x8)	120 views (x6)	180 views (x4)	240 views (x3)	360 views (x2)
FBP	28.9182	32.0717	33.8028	38.2559	40.7448	45.4611
U-Net	40.3733	42.1512	43.6840	44.9418	46.4402	47.5937
Dual frame U-Net	40.3775	42.1462	43.6973	44.9717	46.4653	47.6765
Tight frame U-Net	40.4856	42.2380	43.7682	45.0406	46.4847	47.5797

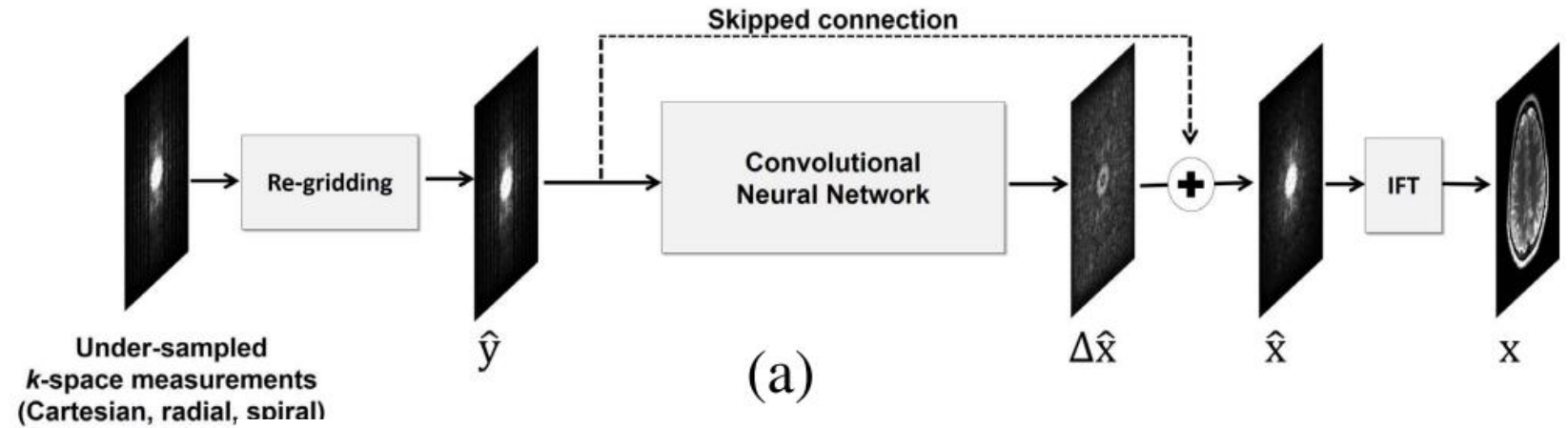
[2] Yoseb Han and Jong Chul Ye, Framing U-Net via Deep Convolutional Framelets: Application to Sparse-View CT, TMI 2018



Solve Pb in Sensor domain

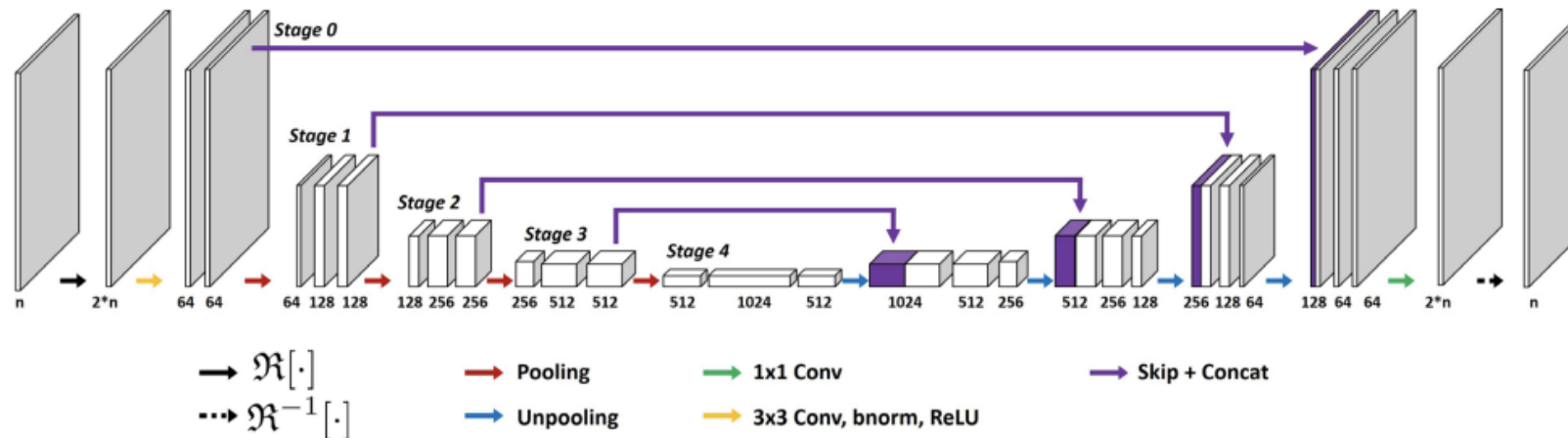
[1] Yoseo Han, Leonard Sunwoo , Jong Chul Ye: k-Space Deep Learning for Accelerated MRI, TMI 2020

PB Formulation 2: Solve Pb in sensor domain



$$\min_{\hat{\mathbf{z}} \in \mathbb{C}^N} \quad \left\| x - \mathcal{F}^{-1}[\hat{\mathbf{z}}] \right\|^2$$

subject to $\text{RANK } \mathbb{H}_d(\hat{\mathbf{z}}) = s, \quad \mathcal{P}_\Lambda[\hat{\mathbf{x}}] = \mathcal{P}_\Lambda[\hat{\mathbf{z}}]$

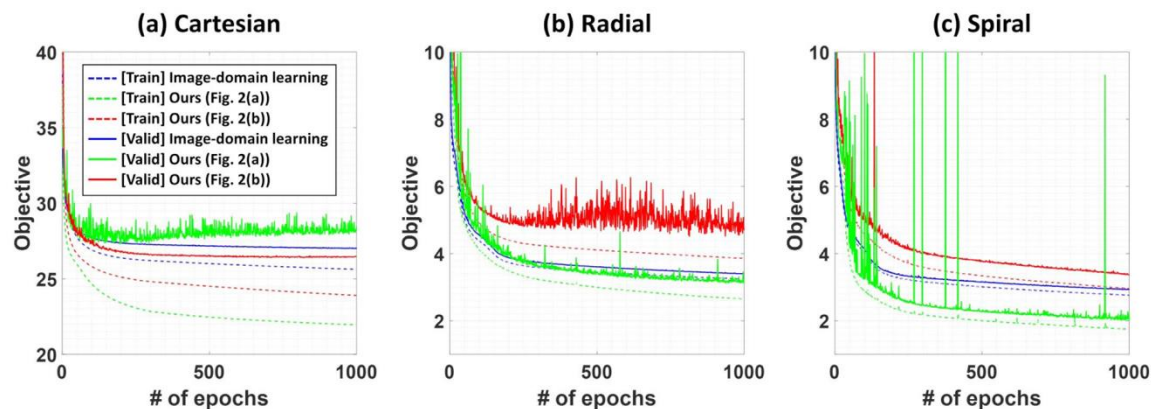
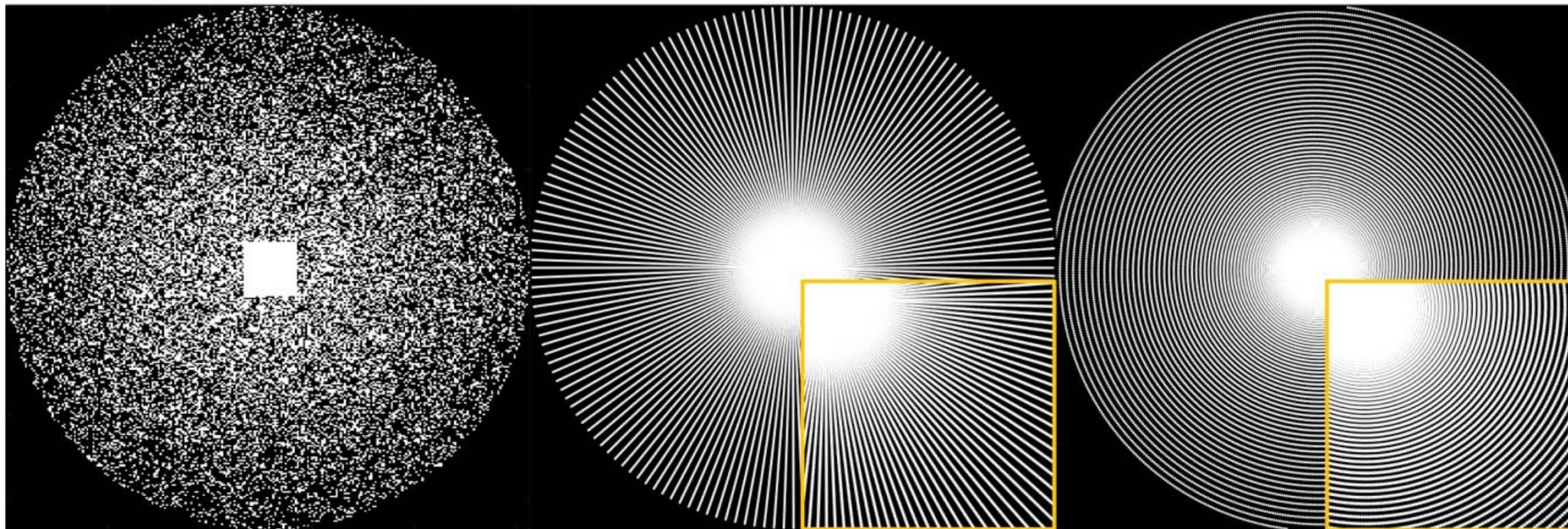


PB Formulation 2: Solve Pb in sensor domain

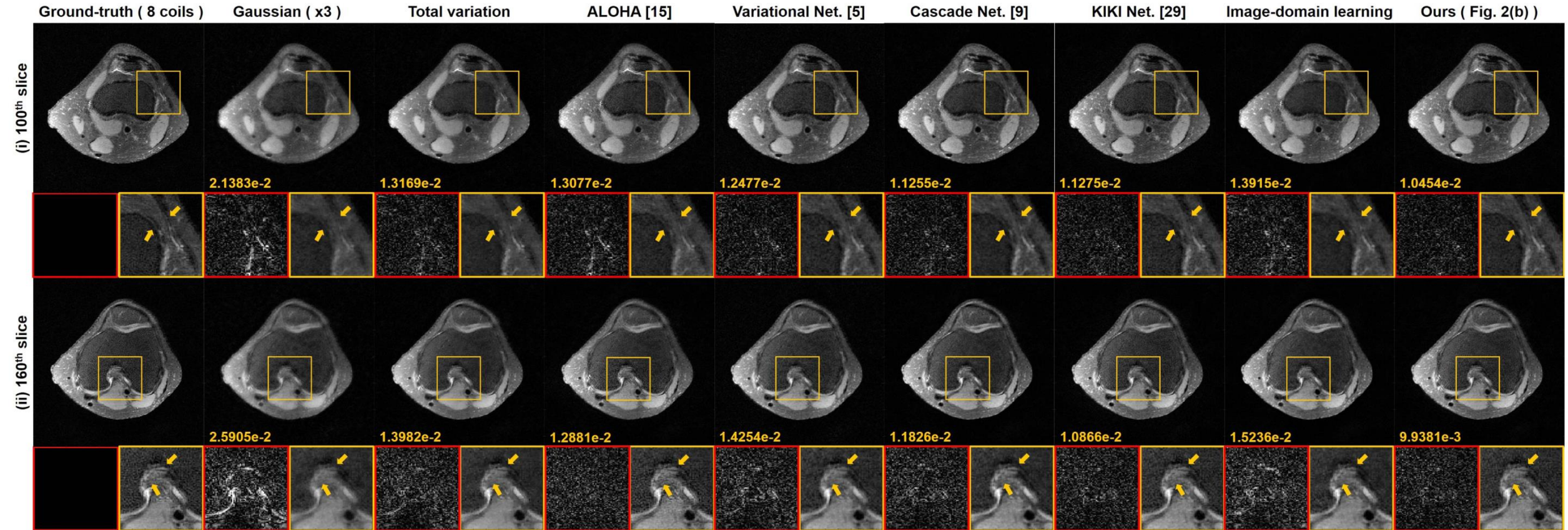
(a) Cartesian

(b) Radial

(c) Spiral



PB Formulation 2: Solve Pb in sensor domain



QUANTITATIVE COMPARISON FROM CARTESIAN TRAJECTORY AT $R = 3$ IN 8 COILS PARALLEL IMAGING

Metric	Input (8 coils, $\times 3$)	Total variation	ALOHA [15]	Variational Net. [5]	Cascade Net. [9]	KIKI Net. [29]	Image-domain learning	Ours (Fig. 3(b))
PSNR [dB]	34.2021	36.0689	36.1013	36.1428	36.6055	36.6847	35.8497	36.9931
NMSE ($\times 10^{-2}$)	2.3384	1.5826	1.5812	1.5379	1.4196	1.4086	1.6497	1.3154
SSIM	0.7710	0.7878	0.7914	0.7966	0.7982	0.8018	0.7776	0.8087
Times (sec/slice)	-	1.0938	16.5554	0.1688	0.2219	0.5438	0.1188	0.1438

Solve Pb in both domain

- [1] Burhaneddin Yaman, Seyed Amir Hossein Hosseini, Steen Moeller, Jutta Ellermann, Kâmil Uğurbil, Mehmet Akçakaya: Self-supervised learning of physics-guided reconstruction neural networks without fully sampled reference data, MRM 2019
- [2] Ce Wang, Kun Shang, Haimiao Zhang, Qian Li, Yuan Hui, S. Kevin Zhou
DuDoTrans: Dual-Domain Transformer Provides More Attention for Sinogram Restoration in Sparse-View CT Reconstruction, CVPR 2019; TMI 2021

PB Formulation 3: Solve Pb in both domain

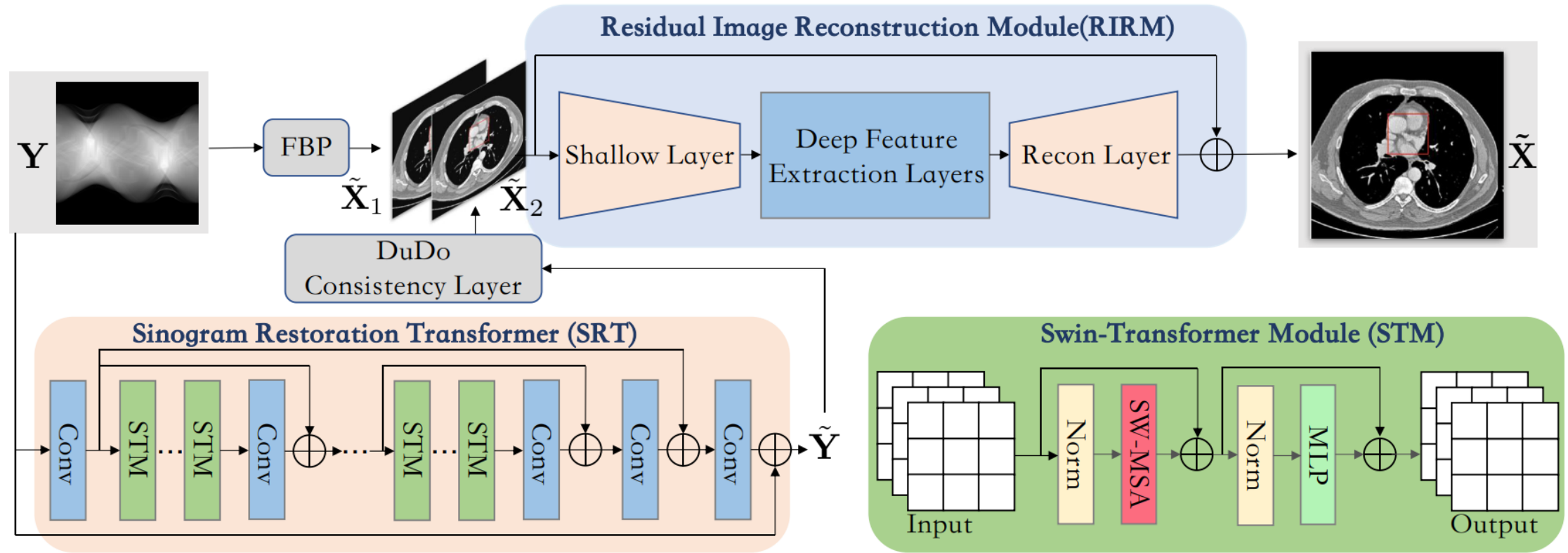


Figure 2. The framework of proposed DuDoTrans for SV CT image reconstruction. When under-sampled sinograms are given, our DuDoTrans first restores clean sinograms with SRT, followed by RIRM to reconstruct the CT image with both restored and raw sinograms.

PB Formulation 3: Solve Pb in both domain

$$\arg \min_x \left\| \mathbf{y}_\Omega - \mathbf{E}_\Omega \mathbf{x} \right\|_2^2 + \mathcal{R}(\mathbf{x}), \quad (2)$$

$$\arg \min_{x,z} \left\| \mathbf{y}_\Omega - \mathbf{E}_\Omega \mathbf{x} \right\|_2^2 + \mu \left\| \mathbf{x} - \mathbf{z} \right\|_2^2 + \mathcal{R}(\mathbf{z}), \quad (3)$$

$$\mathbf{z}^{(i-1)} = \arg \min_z \mu \left\| \mathbf{x}^{(i-1)} - \mathbf{z} \right\|_2^2 + \mathcal{R}(\mathbf{z}), \quad (4)$$

$$\mathbf{x}^{(i)} = \arg \min_x \left\| \mathbf{y}_\Omega - \mathbf{E}_\Omega \mathbf{x} \right\|_2^2 + \mu \left\| \mathbf{x} - \mathbf{z}^{(i-1)} \right\|_2^2, \quad (5)$$

$$\mathbf{x}^{(i)} = \left(\mathbf{E}_\Omega^H \mathbf{E}_\Omega + \mu \mathbf{I} \right)^{-1} \left(\mathbf{E}_\Omega^H \mathbf{y} + \mu \mathbf{z}^{(i-1)} \right), \quad (6)$$



PB Formulation 3: Solve Pb in both domain

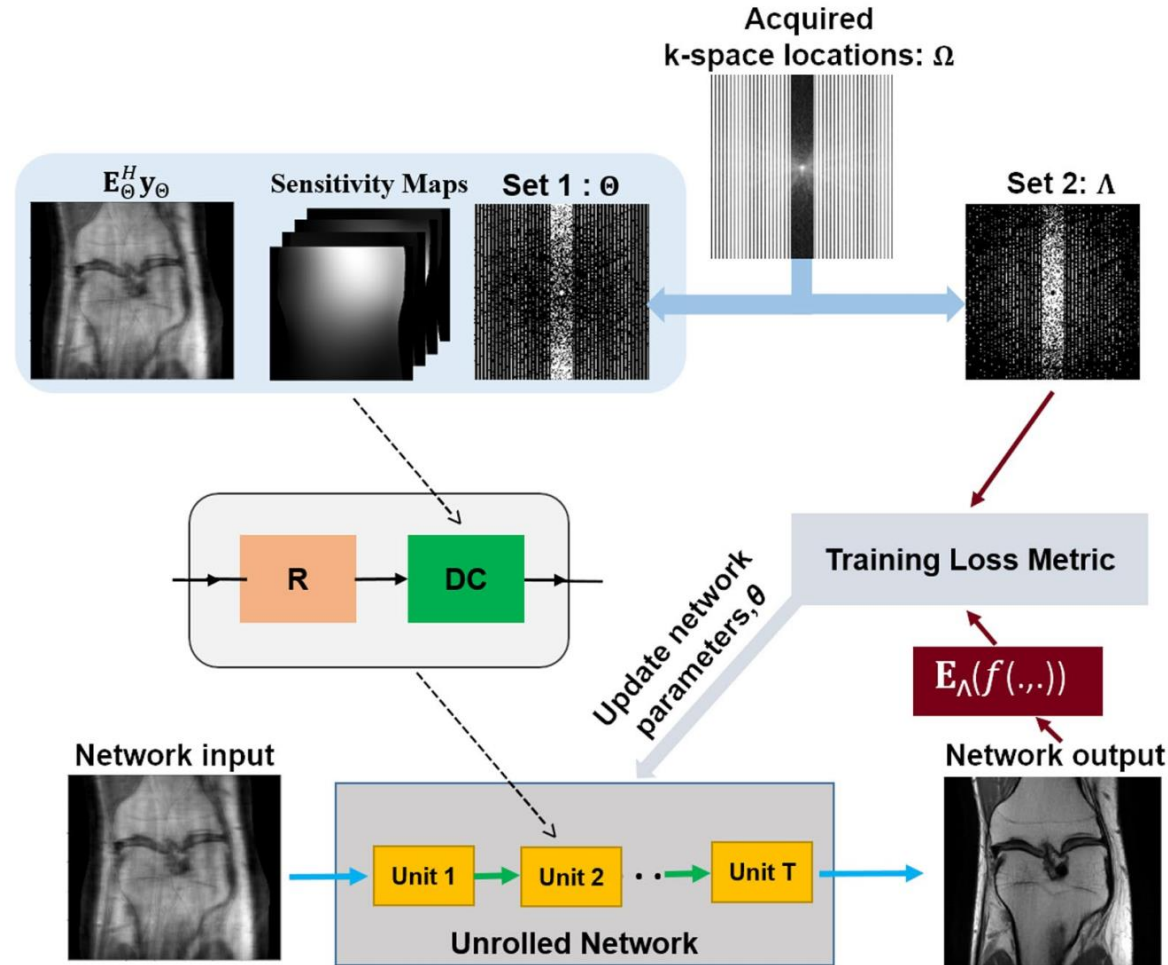
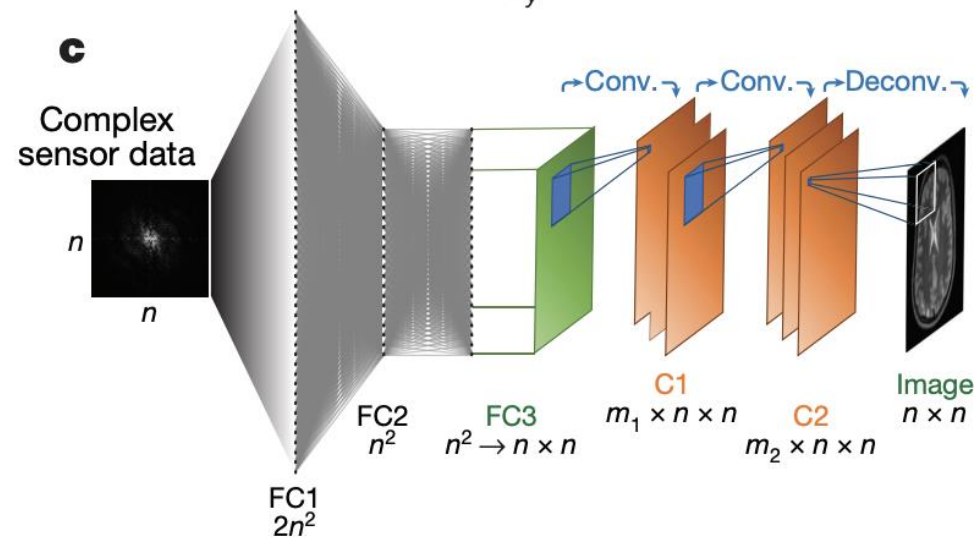
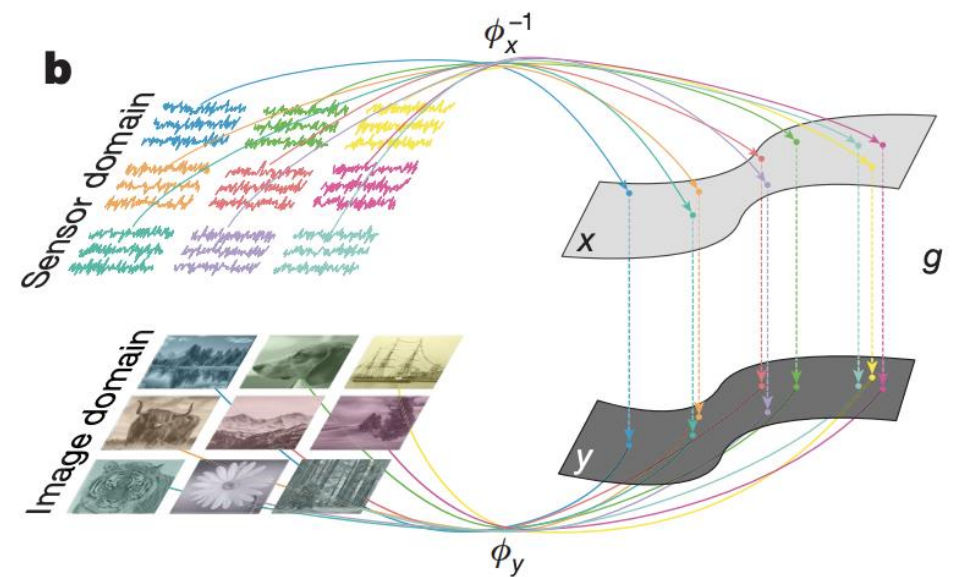
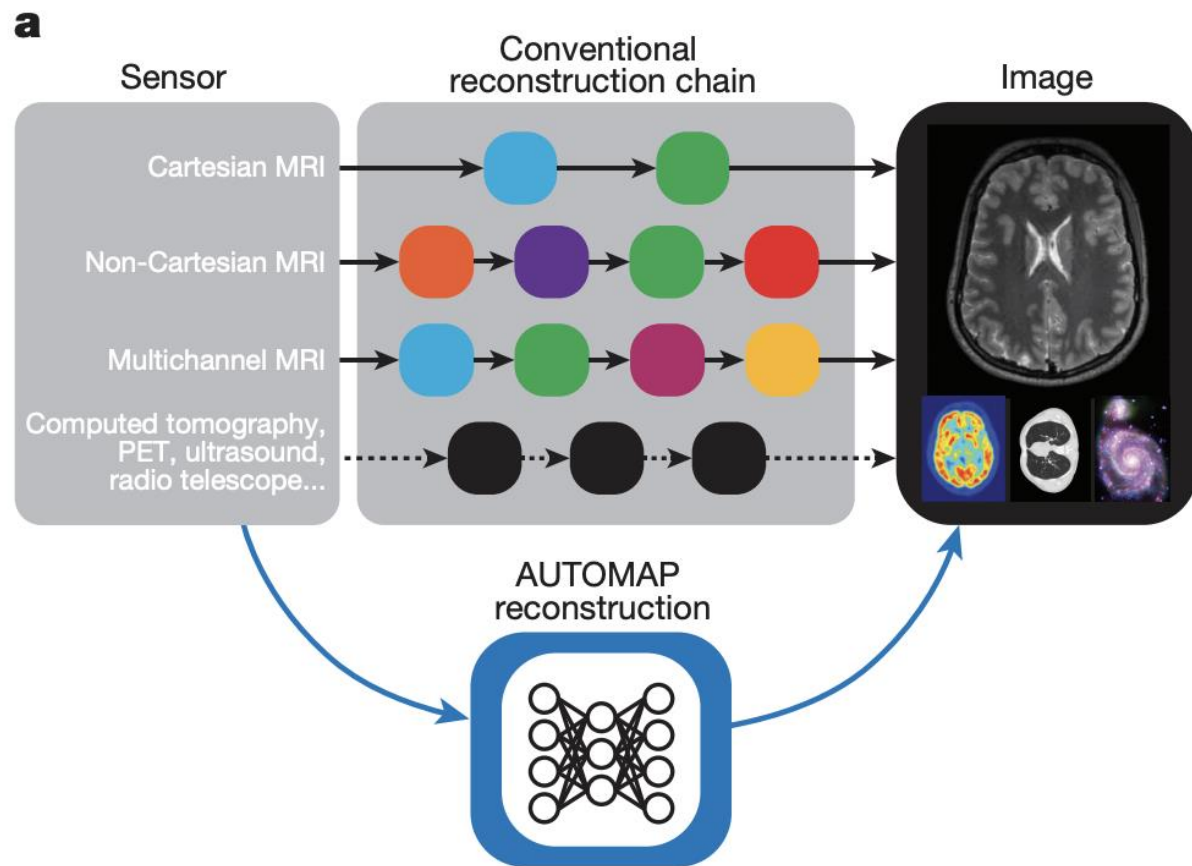


FIGURE 2 The self-supervised learning scheme to train physics-guided deep learning without fully sampled data. The acquired subsampled k-space measurements, Ω , are split into two disjoint sets, Θ and Λ . The first set of indices, Θ , is used in the data consistency unit of the unrolled network, whereas the latter set, Λ , is used to define the loss function for training. During training, the output of the network is transformed to k-space, and the available subset of measurements at Λ are compared with the corresponding reconstructed k-space values. Based on this training loss, the network parameters are subsequently updated

Learning Transform in-between Domain

[1] Bo Zhu, Jeremiah Z. Liu, Stephen F. cauley, Bruce r. rosen & matthew S. rosen: Image reconstruction by domain-transform manifold learning , Nature 2018



Take home message

- CT and MRI are relatively ideal background in image reconstruction tasks.
- Most academic works are based on simulated data, real imaging tasks are much more complex.
- Comparing with deep learning network architecture, it is more important to think about task formulation.
- Devils are always in details, with right task formulation, performance is highly dependent on implementation details.

